ABDELWAHEB HAFS

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Education

Université Paris-Saclay — **Paris, France**
Ph.D. in Sport and Human Movement Sciences — *2021-2025*
- **Dissertation:** *Game-theoretic predictive control for intuitive movement assistance in human-robot interaction*
- **Award:** Demeny-Vaucanson Prize, 2022

Université Paris-Saclay — **Paris, France**
Master's Degree in Robotics, Assistance, and Mobility (RAM) — *2020-2021*
- **Master's thesis:** Adaptive Control Laws for Improving Human–Exoskeleton Interaction
- **Rank:** 1st out of 10 students

National Polytechnic School — **Oran, Algeria**
Bachelor's and Master's Degrees in Automatic Control — *2017-2020*
- **Master's thesis:** Predictive Control Applied to a Two-Wheeled Mobile Robot
- **Rank:** 2nd out of 29 students

School of Electrical and Energetic Engineering of Oran — **Oran, Algeria**
Preparatory Classes — *2015-2017*
- **Rank:** 3rd out of 395 students

Academic Positions

Faculty of Sport Sciences, Université Paris-Saclay — **Paris, France**
Temporary Teaching and Research Associate (ATER) — *Sep. 2025-Aug. 2026*
- **Research:** Implementation of intuitive assistive control laws with trajectory prediction for the study of human–robot interaction, in collaboration with CIAMS and INRIA
- **Teaching:** 192 hours in Biomechanics (L1 STAPS), Robotics (M1 ISMH), and Programming with Python (M2 ISMH)

CIAMS, Université Paris-Saclay — **Paris, France**
Ph.D. Researcher — *Oct. 2021-Dec. 2025*
- Conducted research on game-theoretic and predictive control for human–robot interaction
- Supervised undergraduate/master's interns involved in research projects
- Taught 192 hours in Biomechanics, Robotics, and Programming

LURPA (ENS Paris-Saclay) and CIAMS (Université Paris-Saclay) — **Paris, France**
Master's Research Internship — *Mar. 2021-Sep. 2021*
- Developed a control system based on adaptive oscillators
- Identified the dynamics of a coupled system
- Used Qualisys Track Manager for motion analysis experiments

Skills

Programming and Scientific Computing: MATLAB, Python, C++, C, LaTeX
Software and Tools: SolidWorks, LabVIEW, TIA Portal, VHDL, Inkscape, Proteus Professional
Languages: Arabic, French, English

Publications

- Hafs, A., Verdel, D., Jerray, J., Bruneau, O., Vignais, N., Berret, B., & Fribourg, L. (2023). Convergence and robustness of the Hopf oscillator applied to an ABLE exoskeleton: reachability analysis and experimentation. *Modélisation des Systèmes Réactifs (MSR'23)*, Toulouse, France.
- Hafs, A., Verdel, D., Burdet, E., Bruneau, O., & Berret, B. (2024). Finite-horizon inverse differential game approach for optimal trajectory-tracking assistance with a wrist exoskeleton. *BioRob 2024.* DOI: 10.1109/BioRob60516.2024.10719810
- Hafs, A., Verdel, D., Bruneau, O., & Berret, B. (2025). Game-theoretic interaction control for assistive exoskeletons: a 2-DOF simulation study. *IFAC-PapersOnLine.* DOI: 10.1016/j.ifacol.2025.10.231
- Hafs, A., Farr, A., Verdel, D., Bruneau, O., Burdet, E., & Berret, B. (2026). Model predictive game control for personalized and targeted interactive assistance. *Communications Engineering.* DOI: 10.1038/s44172-026-00605-8
- Hafs, A. (2025). *Game-theoretic predictive control for intuitive movement assistance in human–robot interaction.* Ph.D. dissertation, Université Paris-Saclay.

Conference and Workshop Communications

- Hafs, A., Verdel, D., Gomes, W., Bruneau, O., & Berret, B. (2023). Optimizing human–robot interactions through differential game control. Workshop on Multilimb Coordination in Human Neuroscience and Robotics: Classical and Learning Perspectives, IROS 2023, Detroit, USA.
- Société Francophone Posture Équilibre et Locomotion (SOFPEL), 2024, Paris, France.
- GDR Day on "Contrôle sensori-moteur pour la robotique", 2025, Paris, France.
- Biomeca@Inria Day, 2026, Paris, France.

Internship Supervision

- **Spring 2023** Supervision of an internship on the comparative study of Model Predictive Control and differential game-based control for human-exoskeleton interaction.
- **Feb.-Jul. 2024** Supervision of an internship on the analysis of collaborative human-exoskeleton interaction based on differential game theory.
- **Spring 2025** Supervision of an internship on the analysis of the planning horizon in collaborative human-exoskeleton interaction based on differential game theory.

Reviewing Activity

- Reviewed one abstract for IFAC 2025.